

$$R_{total} = R_{on10} || R_{on11}$$

$$= \frac{1}{\mu_n C_{ox} \frac{W_{10}}{L_{10}} (V_{out1} - V_{th})} || \frac{1}{\mu_n C_{ox} \frac{W_{11}}{L_{11}} (V_{out1} - V_{th})}$$

$$= \frac{1}{\mu_n C_{ox} \frac{W_{10,11}}{L_{10,11}} (V_{out1} + V_{out2} - 2V_{th})}$$

deep triode region

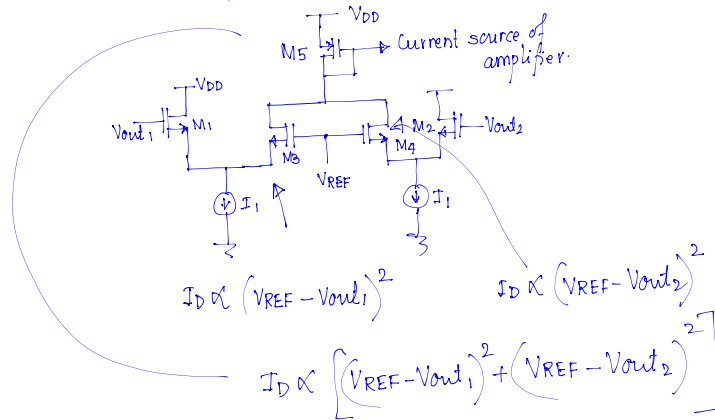
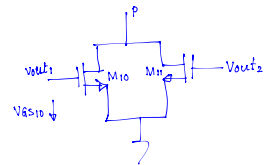
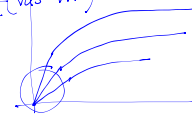
$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} [2(V_{GS} - V_{th})V_{DS} - V_{DS}^2]$$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} 2(V_{GS} - V_{th})V_{DS}$$

$$= \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})V_{DS}$$

$$R_{on} = \frac{V_{DS}}{I_D} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})}$$

$$V_{CCS} \leftarrow I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$$



Feedback mechanism and comparison

